

# Reproduce-then-Extend — Pinyon Jay Habitat Selection (Ecology)

## (Day 2)

Intro to Machine Learning · Bruce A. Desmarais

July 07, 2026

**Published study.** Daw, S. K. et al. (2022). “Behavior-specific occurrence patterns of Pinyon Jays.” *PLoS ONE* (doi:10.1371/journal.pone.0237621; S1 Dataset, open). A **logistic resource-selection function (RSF)** — habitat *use* points (1) vs *available* points (0) on environmental predictors. Use is the minority class, so out-of-sample skill is scored by **AUC-PR** (area under the precision-recall curve).

## 1 Background

Pinyon Jays are a **declining bird** of the southwestern United States, and managers need to know **which habitat they use** in order to conserve it. Daw and colleagues fit a **resource-selection function** – a logistic regression comparing locations the jays *used* to *available* background locations – to identify the environmental conditions associated with jay occurrence. The **primary conclusion** is that jays select particular pinyon-juniper conditions – more open stands, sites near habitat edges, and areas with downed woody debris – which the regression translates into a picture of suitable habitat for management.

## 2 Data and codebook

**Unit of analysis:** a sampled ground location – an observed jay-**use** point vs an **available** (background) point;  $n = 466$  (124 use).

| Variable                      | Definition                                                                      |
|-------------------------------|---------------------------------------------------------------------------------|
| use                           | 1 = observed Pinyon Jay use location, 0 = available location ( <b>outcome</b> ) |
| Elevation..feet.              | elevation (feet)                                                                |
| Slope..percent.               | ground slope (%)                                                                |
| Stand_Age..years.             | forest stand age (years)                                                        |
| Stand_Density_Index           | stand density index (tree density / competition)                                |
| Canopy_Cover..percent.        | overstory canopy cover (%)                                                      |
| Tree_Cover..percent.          | tree cover (%)                                                                  |
| Shrub_Cover..percent.         | shrub cover (%)                                                                 |
| Forb_Cover..percent.          | forb (herbaceous) cover (%)                                                     |
| Grass_Cover..percent.         | grass cover (%)                                                                 |
| Woody_Debris..category.1.-.3. | counts of downed woody debris by size class (1-3)                               |
| Distance.to.Road..km.         | distance to the nearest road (km)                                               |
| DisturbanceYes                | 1 if any recent disturbance was recorded at the site                            |
| Distance.to.Edge..m.          | distance to the nearest habitat edge (m)                                        |

### 3 Descriptive analysis

```
d <-
  ↪ read.csv("https://raw.githubusercontent.com/bdesmarais/intro-ml-statistical-horizons-4day/main/tutorial/01_data/01_data.csv")

# Predictors: keep the interpretable environmental variables and drop the sparse
# one-hot habitat/disturbance-type dummies (unidentified in the logit).
preds <- setdiff(names(d), "use")
preds <- preds[sapply(d[preds], function(x) length(unique(x)) > 1)]
preds <- preds[!grepl("Habitat_Type|Disturbance_Type", preds)]

clean <- function(v) gsub("[. _]+", " ", v) # tidy mangled names for labels
short <- function(v) { # compact labels for tables/axes
  v <- gsub("[. _]+", " ", v); v <- gsub("percent", "%", v)
  v <- gsub("category", "cat", v); trimws(v)
}
num <- setdiff(preds, "DisturbanceYes") # 14 numeric environmental vars
```

#### 3.1 Dataset and provenance

The unit of analysis is a **sampled ground location** in pinyon–juniper woodland of the southwestern United States. Each row is either a location where Pinyon Jays were **observed** (a *use* point) or a randomly drawn **available** (background) point that characterizes what habitat was on offer. This use-vs-available design is the standard setup for a **resource-selection function**: the model learns which environmental conditions distinguish used from available locations. The data are the open **S1 Dataset** from Daw et al. (2022, *PLoS ONE*).

There are **466 locations** (124 use, 342 available) and **15 predictors** retained for modeling — 14 continuous environmental measurements plus one binary disturbance indicator. (The raw file also carries sparse one-hot `Habitat_Type*` and `Disturbance_Type*` dummies; these are dropped above because they are near-collinear and unidentified in the logistic RSF.)

Table 1: Variable dictionary: outcome and retained predictors.

| Variable                  | Role      | Type    | Description                          |
|---------------------------|-----------|---------|--------------------------------------|
| use                       | outcome   | binary  | 1 = jay use location, 0 = available  |
| Elevation..feet.          | predictor | numeric | elevation (feet)                     |
| Slope..percent.           | predictor | numeric | ground slope (%)                     |
| Stand_Age..years.         | predictor | numeric | forest stand age (years)             |
| Stand_Density_Index       | predictor | numeric | stand density index (competition)    |
| Canopy_Cover..percent.    | predictor | numeric | overstory canopy cover (%)           |
| Tree_Cover..percent.      | predictor | numeric | tree cover (%)                       |
| Shrub_Cover..percent.     | predictor | numeric | shrub cover (%)                      |
| Forb_Cover..percent.      | predictor | numeric | forb (herb) cover (%)                |
| Grass_Cover..percent.     | predictor | numeric | grass cover (%)                      |
| Woody_Debris..category.1. | predictor | count   | downed wood, small size class        |
| Woody_Debris..category.2. | predictor | count   | downed wood, medium size class       |
| Woody_Debris..category.3. | predictor | count   | downed wood, large size class        |
| Distance.to.Road..km.     | predictor | numeric | distance to nearest road (km)        |
| DisturbanceYes            | predictor | binary  | 1 if recent disturbance recorded     |
| Distance.to.Edge..m.      | predictor | numeric | distance to nearest habitat edge (m) |

### 3.2 The outcome: used vs available

The outcome **use** is **binary and imbalanced**: only **26.6%** of locations are use points, so *available* is the majority class at a **2.8:1** ratio. This base rate is the single most important fact for evaluation. A trivial model that always predicts “available” would already be 73.4% accurate, so **raw accuracy is misleading** here — a good classifier must find the minority *use* class. That is exactly why the tutorial scores held-out skill with **AUC-PR** (area under the precision–recall curve), whose no-skill baseline equals the base rate (0.266), not 0.5.

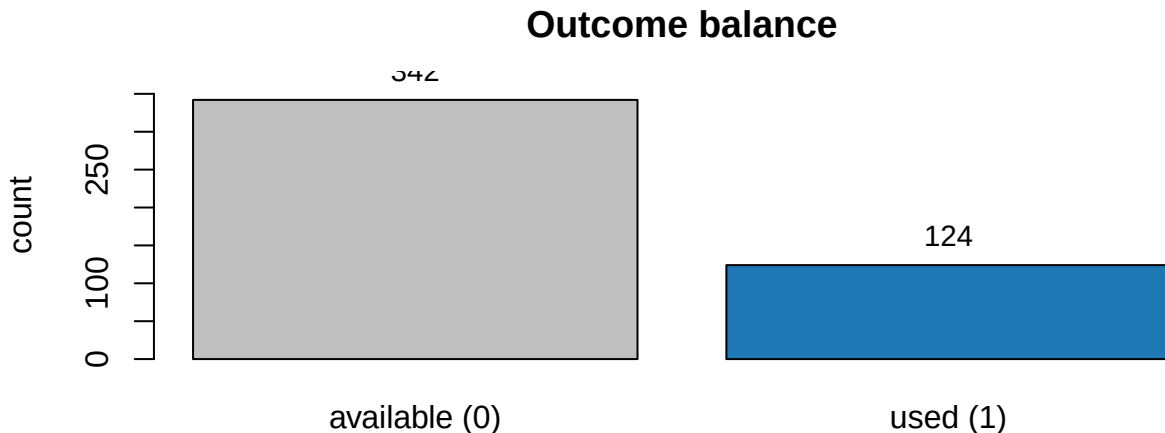


Figure 1: Class balance of the use-vs-available outcome. Use is the minority class.

### 3.3 Univariate predictor distributions

The 14 continuous predictors span very different scales (elevation in thousands of feet, cover in percent, debris as small counts), so we summarize them in a single table with location, spread, quartiles and **skewness**, then show the shapes as faceted histograms.

Table 2: Univariate summary of the numeric predictors.

|                     | n   | miss | mean   | sd     | min  | Q1     | med    | Q3     | max     | skew |
|---------------------|-----|------|--------|--------|------|--------|--------|--------|---------|------|
| Elevation feet      | 466 | 0    | 6810.0 | 686.1  | 4430 | 6275.5 | 6760.0 | 7270.0 | 8700.0  | 0.2  |
| Slope %             | 466 | 0    | 28.4   | 17.6   | 0    | 15.0   | 25.0   | 40.0   | 91.0    | 0.7  |
| Stand Age years     | 466 | 0    | 143.2  | 81.2   | 0    | 86.0   | 136.0  | 194.8  | 548.0   | 0.6  |
| Stand Density Index | 466 | 0    | 154.7  | 97.5   | 0    | 82.7   | 146.8  | 220.7  | 542.0   | 0.5  |
| Canopy Cover %      | 466 | 0    | 27.9   | 16.0   | 0    | 17.0   | 28.0   | 40.0   | 74.0    | 0.1  |
| Tree Cover %        | 466 | 0    | 19.1   | 13.4   | 0    | 7.8    | 18.8   | 28.6   | 59.8    | 0.4  |
| Shrub Cover %       | 466 | 0    | 6.1    | 6.0    | 0    | 2.0    | 4.0    | 7.9    | 37.2    | 1.9  |
| Forb Cover %        | 466 | 0    | 1.6    | 2.6    | 0    | 1.0    | 1.0    | 1.2    | 26.0    | 5.9  |
| Grass Cover %       | 466 | 0    | 5.8    | 8.8    | 0    | 1.2    | 2.2    | 5.8    | 59.8    | 3.1  |
| Woody Debris cat 1  | 466 | 0    | 15.3   | 21.4   | 0    | 2.0    | 6.0    | 19.0   | 180.0   | 2.7  |
| Woody Debris cat 2  | 466 | 0    | 2.6    | 4.8    | 0    | 0.0    | 1.0    | 3.0    | 46.0    | 4.5  |
| Woody Debris cat 3  | 466 | 0    | 0.9    | 2.2    | 0    | 0.0    | 0.0    | 1.0    | 20.0    | 4.4  |
| Distance to Road km | 466 | 0    | 6.0    | 1.6    | 1    | 5.0    | 6.0    | 7.0    | 9.0     | -0.9 |
| Distance to Edge m  | 466 | 0    | 1684.2 | 2015.4 | -676 | 315.2  | 1000.5 | 2299.2 | 10800.0 | 1.7  |

Several patterns stand out. The **cover** and **woody-debris** variables are strongly **right-skewed** (skew for forb, grass and the debris counts all exceed 2): most locations have little cover or downed wood, with a long

tail of rich sites. **Elevation** and **canopy cover** are close to symmetric. **Distance.to.Edge.m.** reaches a large maximum ( $1.08 \times 10^4$  m) with a long right tail, and it contains a few **negative values** (min = -676), an artifact of how signed distance-to-edge was computed in the source GIS layer — worth noting but harmless for tree-based learners, which respond only to rank order.

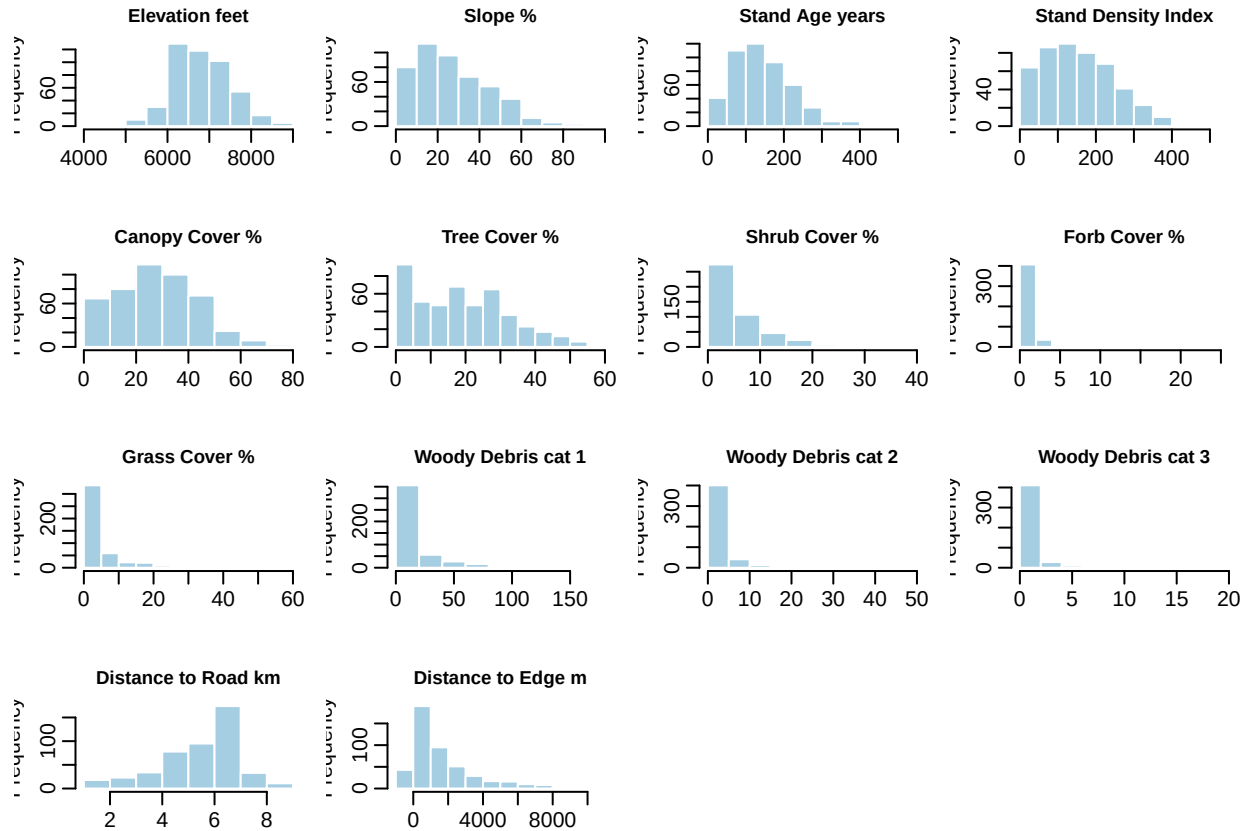


Figure 2: Distributions of the numeric predictors. Cover and woody-debris variables are right-skewed; elevation and canopy are roughly symmetric.

The one binary predictor, **DisturbanceYes**, is recorded at 13.3% of locations (62 of 466) — a minority flag for sites with any recorded recent disturbance.

### 3.4 Missing data

```
## Total missing cells across outcome + predictors: 0
```

```
## Complete cases: 466 of 466 (100%)
```

The analysis matrix is **fully observed** — every outcome and predictor cell is present, so there is no missingness to model and no co-missingness pattern to inspect. The published RSF therefore uses all 466 locations with no imputation or listwise deletion, and the machine-learning comparison inherits the same complete design.

### 3.5 Bivariate association with the outcome

We next ask which conditions actually separate used from available locations. For each numeric predictor we report the **point-biserial correlation** with **use**, the **standardized mean difference** (used minus available, in SDs), and a two-sample *t*-test *p*-value; the table is sorted by the magnitude of the standardized difference.

Table 3: Marginal association of each numeric predictor with use, sorted by |std. mean difference|.

|                     | r      | smd    | p      |
|---------------------|--------|--------|--------|
| Woody Debris cat 1  | 0.422  | 0.955  | <1e-04 |
| Distance to Edge m  | -0.415 | -0.938 | <1e-04 |
| Woody Debris cat 2  | 0.414  | 0.935  | <1e-04 |
| Distance to Road km | -0.394 | -0.891 | <1e-04 |
| Slope %             | -0.381 | -0.862 | <1e-04 |
| Woody Debris cat 3  | 0.320  | 0.723  | <1e-04 |
| Stand Density Index | -0.316 | -0.714 | <1e-04 |
| Elevation feet      | -0.312 | -0.706 | <1e-04 |
| Grass Cover %       | 0.231  | 0.522  | <1e-04 |
| Stand Age years     | -0.204 | -0.462 | <1e-04 |
| Canopy Cover %      | -0.146 | -0.331 | 0.0012 |
| Forb Cover %        | 0.088  | 0.199  | 0.1647 |
| Tree Cover %        | -0.026 | -0.059 | 0.5476 |
| Shrub Cover %       | 0.011  | 0.025  | 0.7980 |

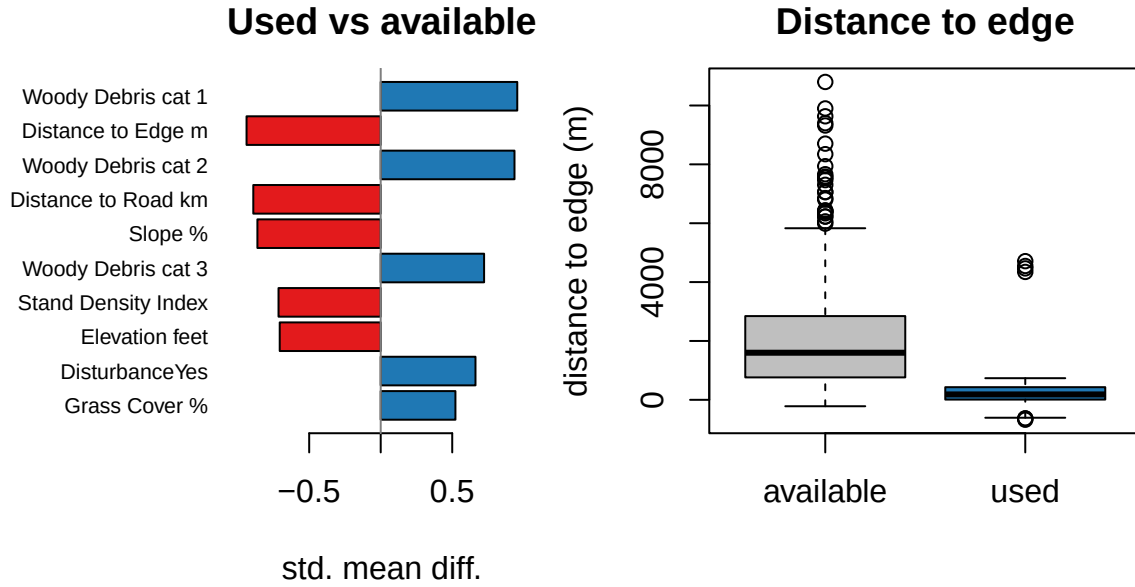


Figure 3: Standardized mean difference (used - available) for each predictor (left) and the strongest gradient, distance-to-edge, by outcome (right).

The signal is clear and ecologically sensible. Use rises with **downed woody debris** (all three size classes positive), and falls sharply with **distance to edge, distance to road, slope, stand density and elevation** — jays select **open, low-density stands near edges with downed wood**, and avoid steep, dense, high-elevation interior sites. Tree and shrub cover show essentially **no marginal association** ( $|r| < 0.03$ ), a hint that their role, if any, is nonlinear or conditional rather than a simple linear gradient. For the binary flag:

```
## Use rate by disturbance flag:
##      0      1
## 0.215 0.597

## chi-square test of independence: p = <1e-04
```

Disturbed sites are used far more often (60% vs 22%), a strong and highly significant association.

### 3.6 Multivariate structure and collinearity

Finally we check whether the predictors are redundant with one another. The heatmap shows pairwise correlations among the numeric predictors, and we compute **variance-inflation factors** from the reproduction model (via auxiliary regressions, since a linear RSF shares the design matrix).

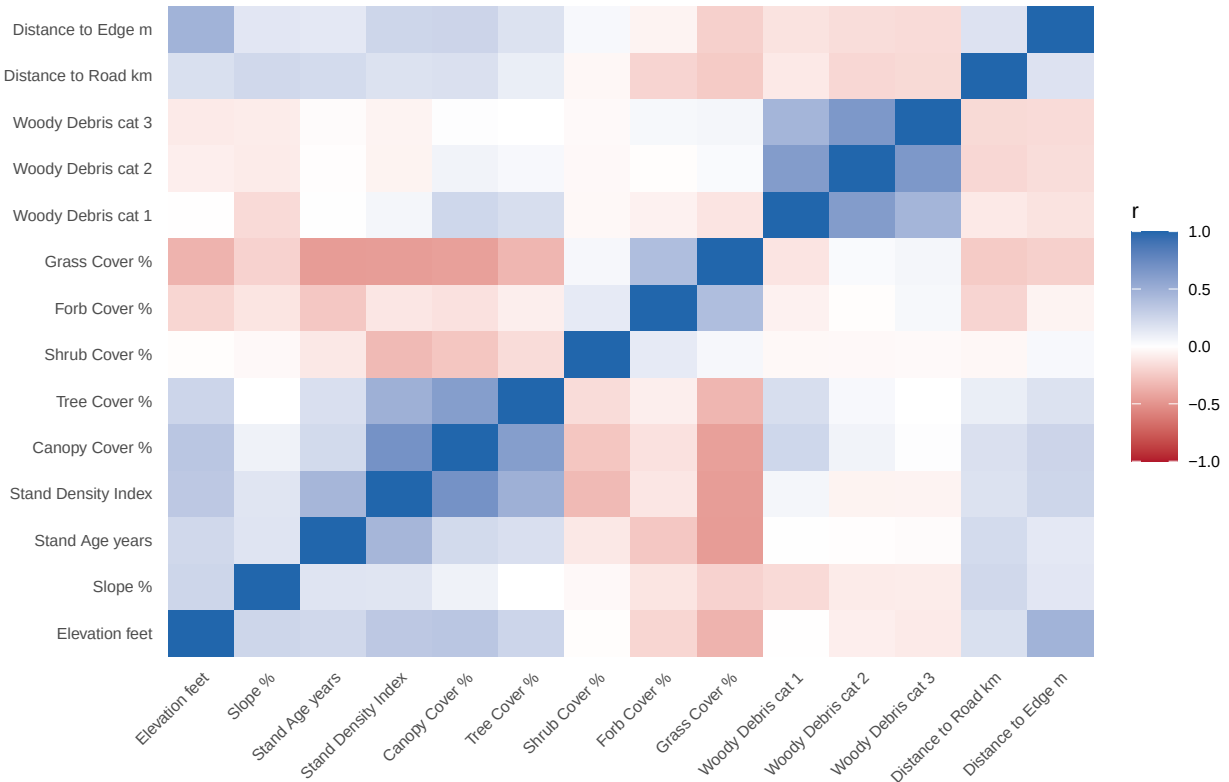


Figure 4: Correlation heatmap of the numeric predictors. No pair exceeds  $|r| = 0.7$ ; the strongest is canopy vs stand-density index.

```
## Largest off-diagonal |r| among numeric predictors: 0.69
## Variance-inflation factors (top 5):
##      Canopy_Cover..percent.      Stand_Density_Index Woody_Debris..category.2.
##              2.78              2.61              2.35
## Woody_Debris..category.1.      Grass_Cover..percent.
##              1.96              1.86
```

Collinearity is **mild**: the strongest correlation is only 0.69 (canopy cover vs stand-density index — both overstory-structure measures), **no pair exceeds**  $|r| = 0.7$ , and the **maximum VIF is 2.78**, far below the usual threshold of 5–10. The predictors therefore carry largely distinct information; the coefficients in the reproduction are individually interpretable, and the modest correlations that do exist (structure, debris size classes) are the kind of interaction-rich texture the tree-based learners can exploit later.

## 4 Reproduce the published RSF regression

The published model is a logistic regression of use vs availability on the environmental predictors.

```
rsf <- glm(reformulate(preds, "use"), data = d, family = binomial)
round(summary(rsf)$coefficients, 3)
```

|                              | Estimate | Std. Error | z value | Pr(> z ) |
|------------------------------|----------|------------|---------|----------|
| ## (Intercept)               | 6.735    | 2.827      | 2.383   | 0.017    |
| ## Elevation..feet.          | 0.000    | 0.000      | -1.080  | 0.280    |
| ## Slope..percent.           | -0.067   | 0.015      | -4.495  | 0.000    |
| ## Stand_Age..years.         | 0.001    | 0.003      | 0.322   | 0.748    |
| ## Stand_Density_Index       | -0.024   | 0.005      | -4.699  | 0.000    |
| ## Canopy_Cover..percent.    | 0.034    | 0.023      | 1.500   | 0.134    |
| ## Tree_Cover..percent.      | 0.080    | 0.026      | 3.108   | 0.002    |
| ## Shrub_Cover..percent.     | -0.015   | 0.035      | -0.435  | 0.664    |
| ## Forb_Cover..percent.      | -0.056   | 0.084      | -0.668  | 0.504    |
| ## Grass_Cover..percent.     | -0.039   | 0.027      | -1.464  | 0.143    |
| ## Woody_Debris..category.1. | 0.029    | 0.014      | 2.132   | 0.033    |
| ## Woody_Debris..category.2. | 0.197    | 0.076      | 2.597   | 0.009    |
| ## Woody_Debris..category.3. | 0.005    | 0.105      | 0.047   | 0.962    |
| ## Distance.to.Road..km.     | -0.452   | 0.134      | -3.383  | 0.001    |
| ## DisturbanceYes            | 2.269    | 0.559      | 4.055   | 0.000    |
| ## Distance.to.Edge..m.      | -0.001   | 0.000      | -4.763  | 0.000    |

**Confirmation against the published study.** The signed coefficients recover the reported habitat-selection pattern — jays select more **open** stands (negative on `Stand_Density_Index` and `Slope..percent.`) and sites **closer to edges** (negative on `Distance.to.Edge..m.`). This logistic RSF is our regression baseline.

## 5 Predictive results: regression vs machine learning

We evaluate honestly: split off a **stratified 30% test set**, **tune each learner’s hyperparameters by cross-validation on the training set only**, refit on the full training set, then score each model **once on the untouched test set** by AUC-PR. Nothing about the test set is used for fitting or tuning.

```
X <- as.matrix(d[, preds]); y <- d$use
aucpr <- function(p, yt) pr.curve(scores.class0 = p[yt == 1], scores.class1 = p[yt ==
  ↪ 0])$auc.integral

set.seed(2026)
te <- c(sample(which(y == 1), round(0.30 * sum(y == 1))),      # stratified 30% test set
  sample(which(y == 0), round(0.30 * sum(y == 0))))
tr <- setdiff(seq_len(nrow(d)), te); yte <- y[te]

## 1. Logistic RSF -- the published regression (no hyperparameters)
m_logit <- glm(use ~ ., data = d[tr, c("use", preds)], family = binomial)
test_auc <- c(Logit = aucpr(predict(m_logit, d[te, ], type = "response"), yte))

## 2. Elastic net -- penalty lambda tuned by CV on the TRAINING set
m_en <- cv.glmnet(X[tr, ], y[tr], alpha = 0.5, family = "binomial")
test_auc["ElasticNet"] <- aucpr(as.numeric(predict(m_en, X[te, ], s = "lambda.min", type
  ↪ = "response")), yte)

## 3. Random forest -- mtry tuned on the TRAINING set by out-of-bag AUC-PR
best_rf <- NULL
for (mt in unique(pmax(1, c(floor(sqrt(ncol(X))), floor(ncol(X) / 3), floor(ncol(X) /
  ↪ 2))))) {
  rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = mt, probability =
  ↪ TRUE)
  sc <- aucpr(rf$predictions[, "1"], y[tr])      # OOB predictions = a training-only
  ↪ CV estimate
  if (is.null(best_rf) || sc > best_rf$sc) best_rf <- list(mtry = mt, sc = sc)
```

```

}
m_rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = best_rf$mtry,
  ↪ probability = TRUE)
test_auc["RandomForest"] <- aucpr(predict(m_rf, X[te, ])$predictions[, "1"], yte)

## 4. XGBoost -- depth x eta tuned by xgb.cv on the TRAINING set; nrounds by early
  ↪ stopping
dtrain <- xgb.DMatrix(X[tr, ], label = y[tr])
best <- NULL
for (depth in c(3, 4, 6)) for (eta in c(0.05, 0.1)) {
  xcv <- xgb.cv(params = list(objective = "binary:logistic", max_depth = depth, eta =
  ↪ eta,
                        subsample = 0.9, colsample_bytree = 0.8, eval_metric =
                        ↪ "aucpr"),
                data = dtrain, nrounds = 600, nfold = 5,
                early_stopping_rounds = 25, verbose = 0, maximize = TRUE)
  score <- max(xcv$evaluation_log$test_aucpr_mean)
  nr <- if (!is.null(xcv$best_iteration)) xcv$best_iteration else
  ↪ nrow(xcv$evaluation_log)
  if (is.null(best) || score > best$score) best <- list(depth = depth, eta = eta, nrounds
  ↪ = nr, score = score)
}
m_xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = best$depth,
  ↪ eta = best$eta,
                        subsample = 0.9, colsample_bytree = 0.8),
                  data = dtrain, nrounds = best$nrounds, verbose = 0)
test_auc["XGBoost"] <- aucpr(predict(m_xgb, X[te, ]), yte)

round(test_auc, 3)

##          Logit   ElasticNet RandomForest      XGBoost
##          0.953       0.970       0.969       0.970

cat(sprintf("Best ML gain over the logistic RSF on the held-out test set: +%.3f
  ↪ AUC-PR\n",
            max(test_auc[-1]) - test_auc["Logit"]))

```

```
## Best ML gain over the logistic RSF on the held-out test set: +0.018 AUC-PR
```

On the held-out test set, random forest and XGBoost lift AUC-PR over the logistic RSF — the nonlinear, interaction-rich structure of habitat selection that a linear-in-the-logit model cannot capture.

## 6 Interpreting the model: importance, SHAP, and partial dependence

We open the fitted learners three ways: **permutation importance**, **SHAP values** (exact TreeSHAP — the signed contribution of each variable to each prediction), and **partial dependence** (the shape of an effect).

```

# Random forest (permutation importance + partial dependence) on all the data
rf <- ranger(x = X, y = factor(y), num.trees = 800, probability = TRUE, importance =
  ↪ "permutation")

# XGBoost fit on all the data, then exact TreeSHAP contributions per observation
xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = 4, eta = 0.05,

```



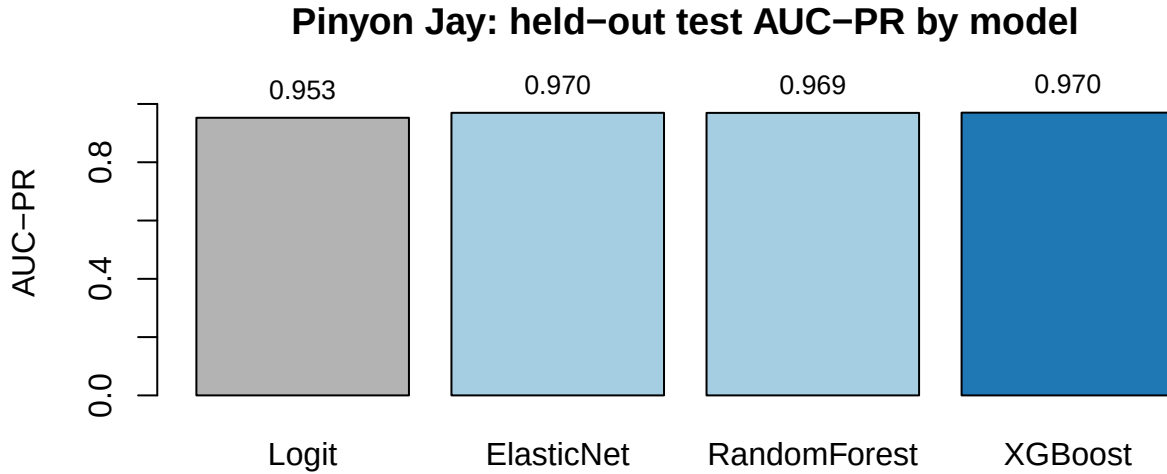


Figure 5: Held-out test-set AUC-PR (higher is better). Grey = published logistic RSF; blue = best ML.

```

subsample = 0.9, colsample_bytree = 0.8),
data = xgb.DMatrix(X, label = y), nrounds = 300, verbose = 0)
shap <- predict(xgb, X, predcontrib = TRUE) # last column is the model bias
shap <- shap[, colnames(X)]               # drop the BIAS column

```

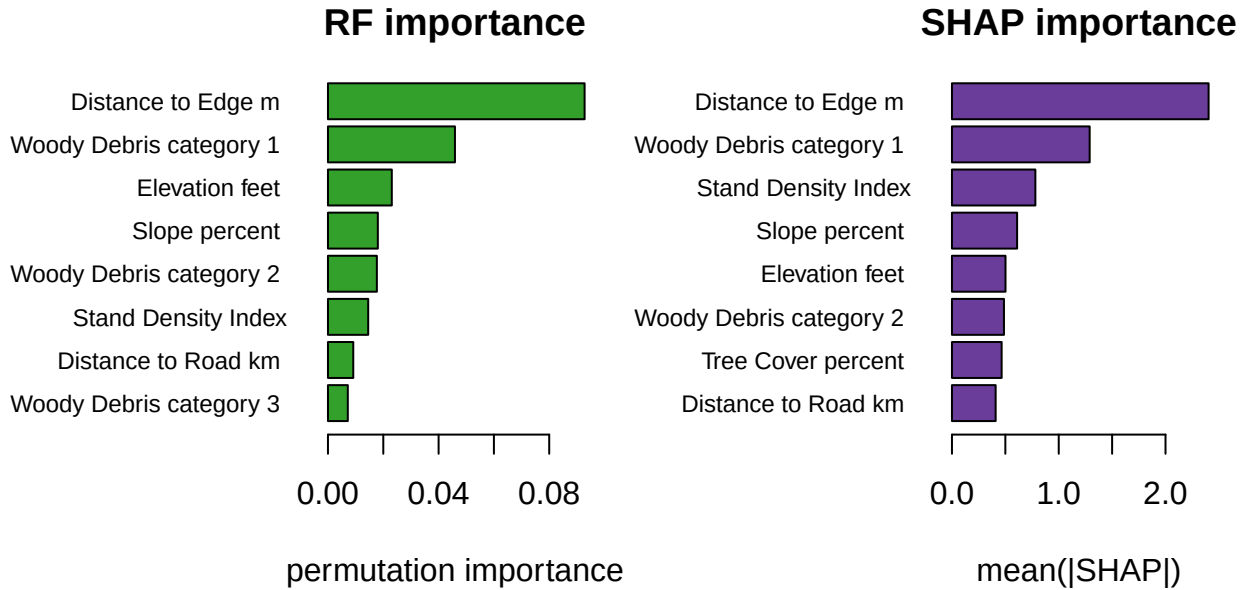


Figure 6: Permutation importance (left) and mean-absolute-SHAP importance (right). The two rankings agree.

**Interpreting the relationships.** The SHAP summary reads directly as habitat selection: use rises as **stand-density index falls** (open blue points sit at positive SHAP, dense red at negative) and **closer to edges**, with **woody debris** present. In plain terms, pinyon jays select **open pinyon-juniper woodland near edges with downed wood** — and avoid dense, closed stands.

Both curves show a strongly **nonlinear** effect — the contribution turns over across the range rather than following one logit slope, which (with the interactions SHAP captures locally) is why the learners beat the linear RSF.

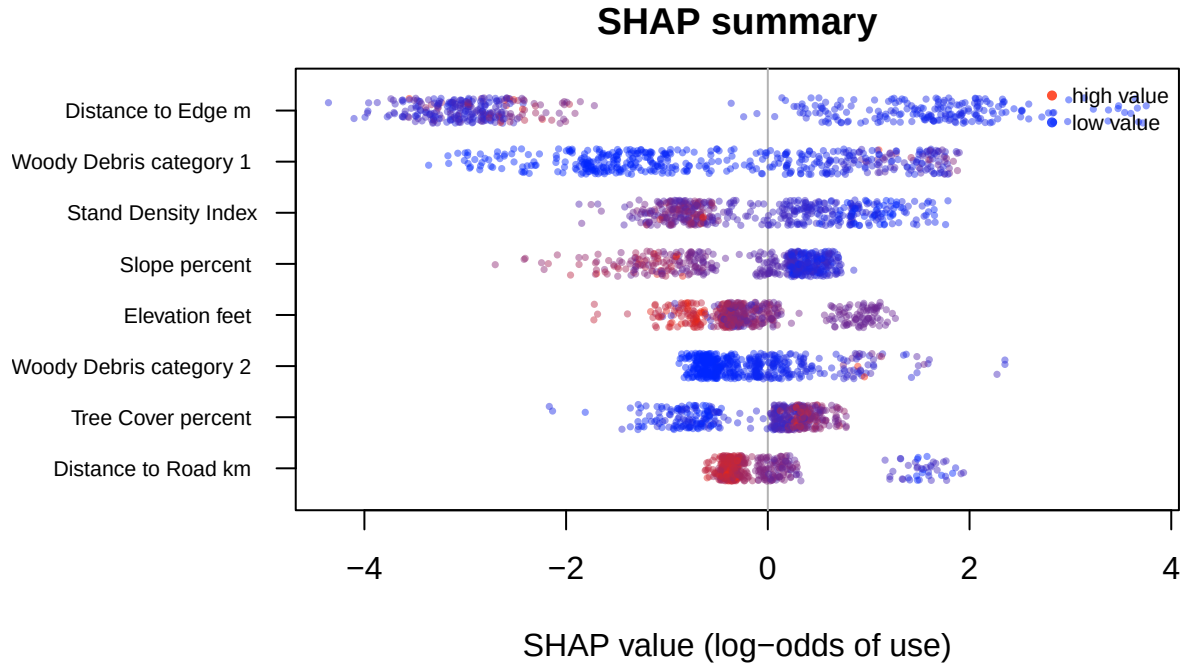


Figure 7: SHAP summary. Each point is one location; x = its SHAP contribution to the log-odds of use; colour = the variable's value (blue low, red high).

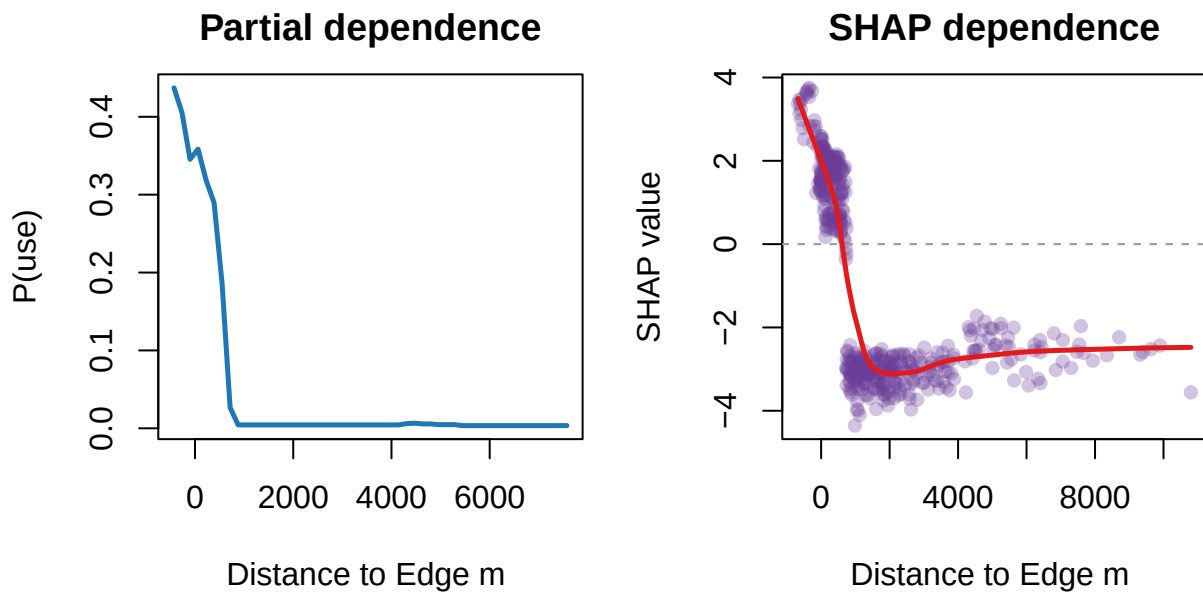


Figure 8: Partial dependence (left) and SHAP dependence (right) for the most influential variable.

## 7 Takeaway

Birds select habitat at nonlinear optima and via interactions — where the linear RSF struggles and trees excel — and SHAP lets us read those relationships (open stands, edges, woody debris) directly off the fitted model. The first of two ecology cases showing ML both predicts better *and* interprets cleanly.

## 8 Recommended exercises

1. Vary the random-forest hyperparameters (number of trees, features per split) and report the effect on AUC-PR.
2. Generate a partial-dependence plot for a second habitat variable and describe the nonlinearity.
3. Compare RF vs XGBoost AUC-PR on the same folds; which wins, and by how much?